A comprehensive study of SWOT PIXC data for flood extent extraction

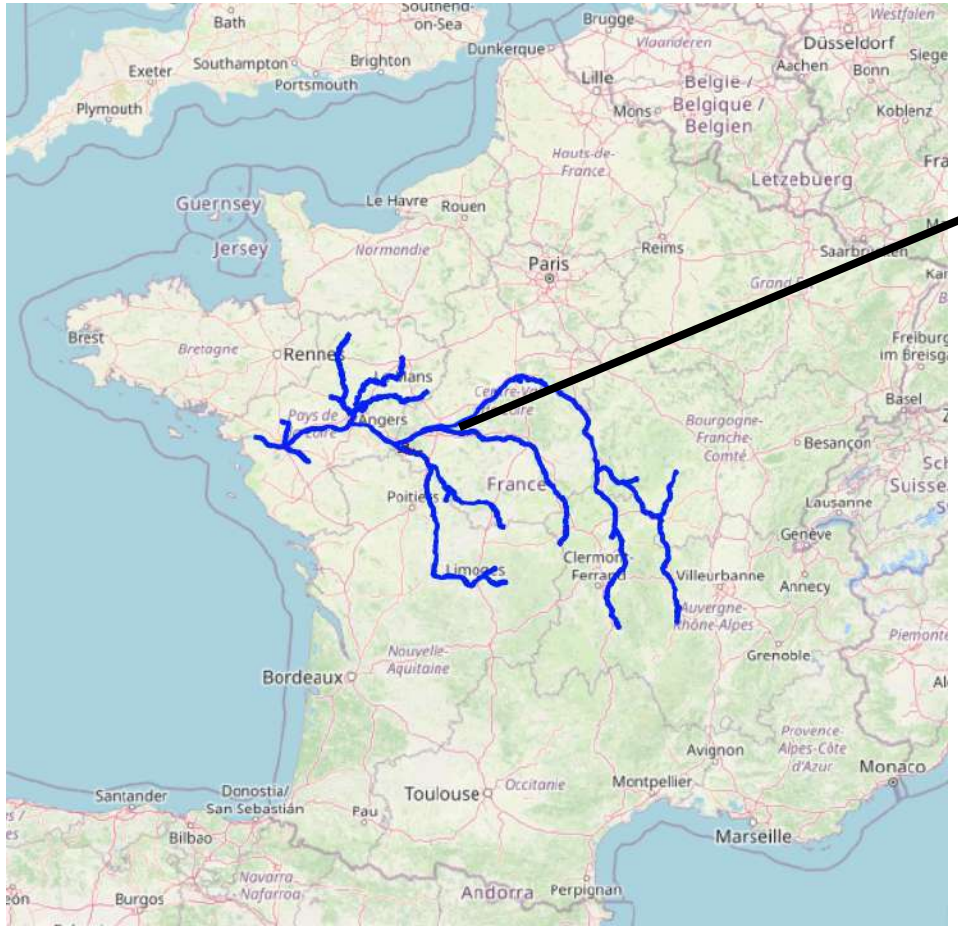
Quentin Bonassies(*), C. Fatras, S. Pena Luque, P. Dubois, L. Cassan, A. Piacentini, S. Ricci, T-H. Nguyen

* contact: bonassies@cerfacs.fr



Context

Wine production area in the Loire valley, close to Tours: Chinon!

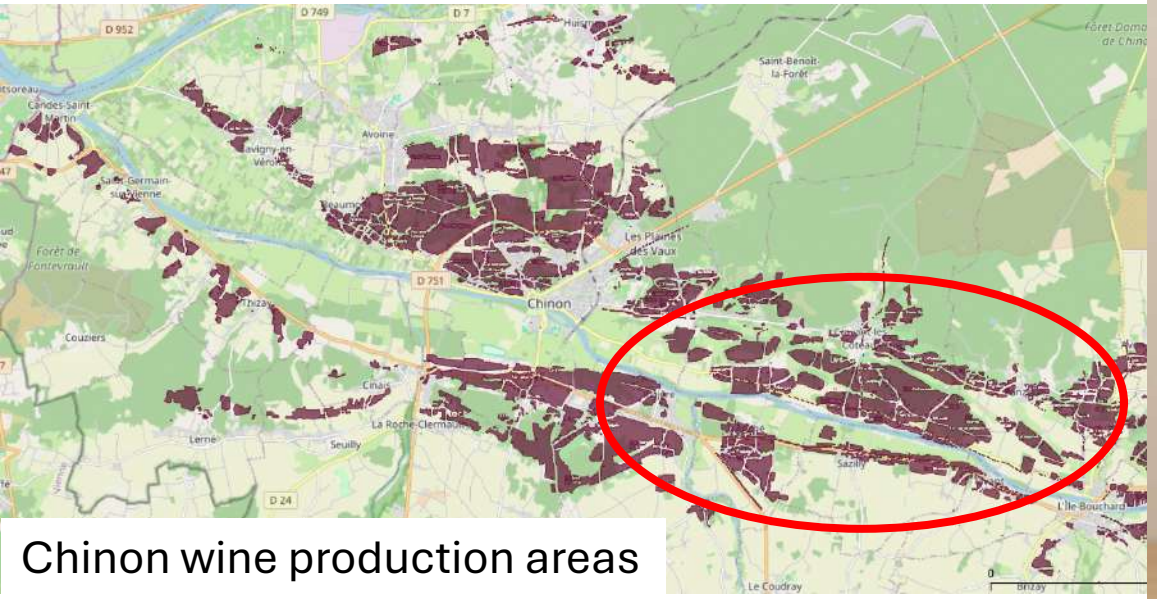






Context

Vienne river flood… on March 31st, 2024



Chinon wine production areas



As observed by Sentinel-1 (VV)



17h49 UTC

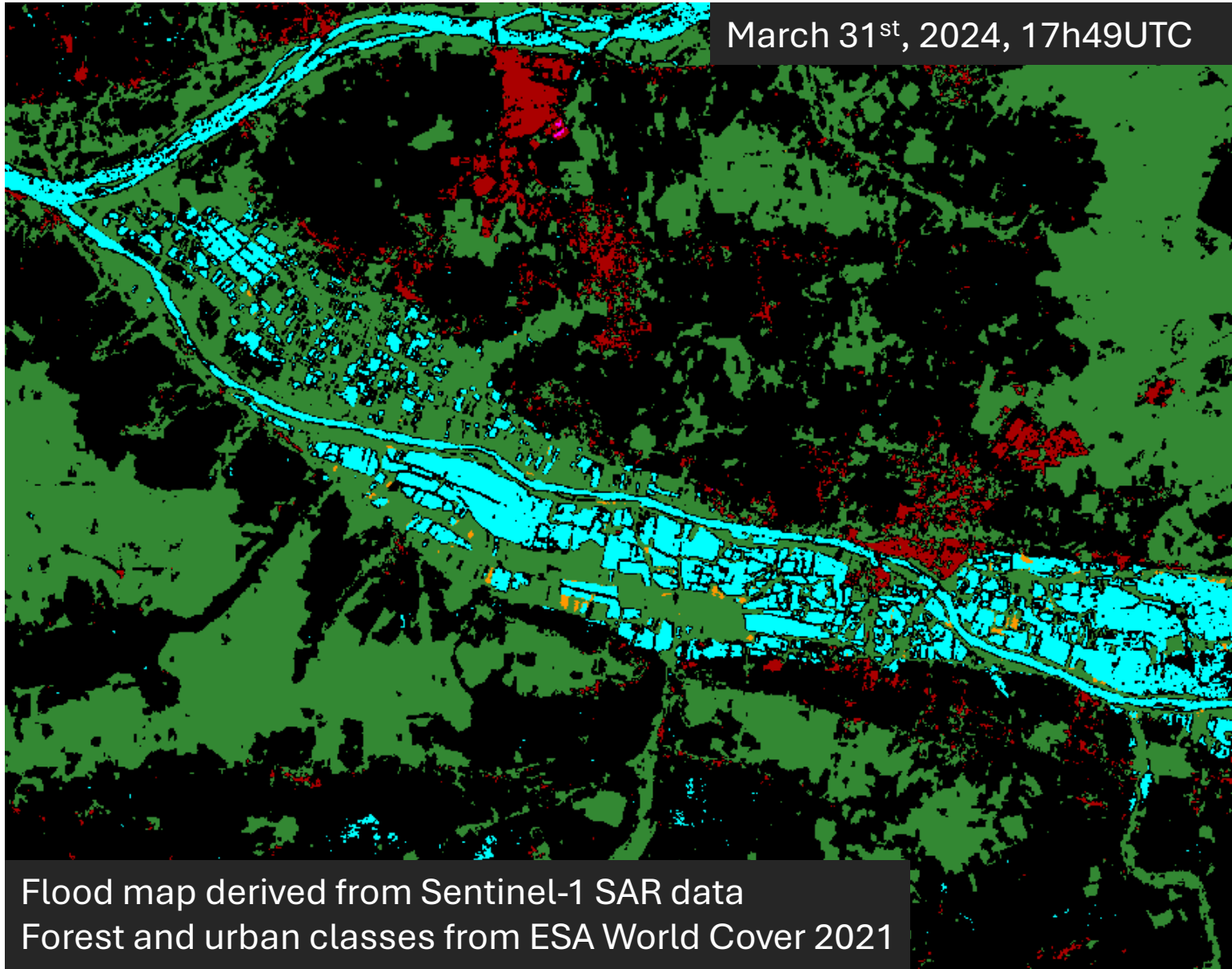
Vineyards clearly affected by flooding, but the rest…..

Context

What is the issue with traditional flood mapping?

- Flooded forest cannot be observed directly
- Flooded urban areas are not « easily » detected
- Combination with DEM powerful, but relies strongly on the quality of all ancillary data

By curiosity, is there any close SWOT data?



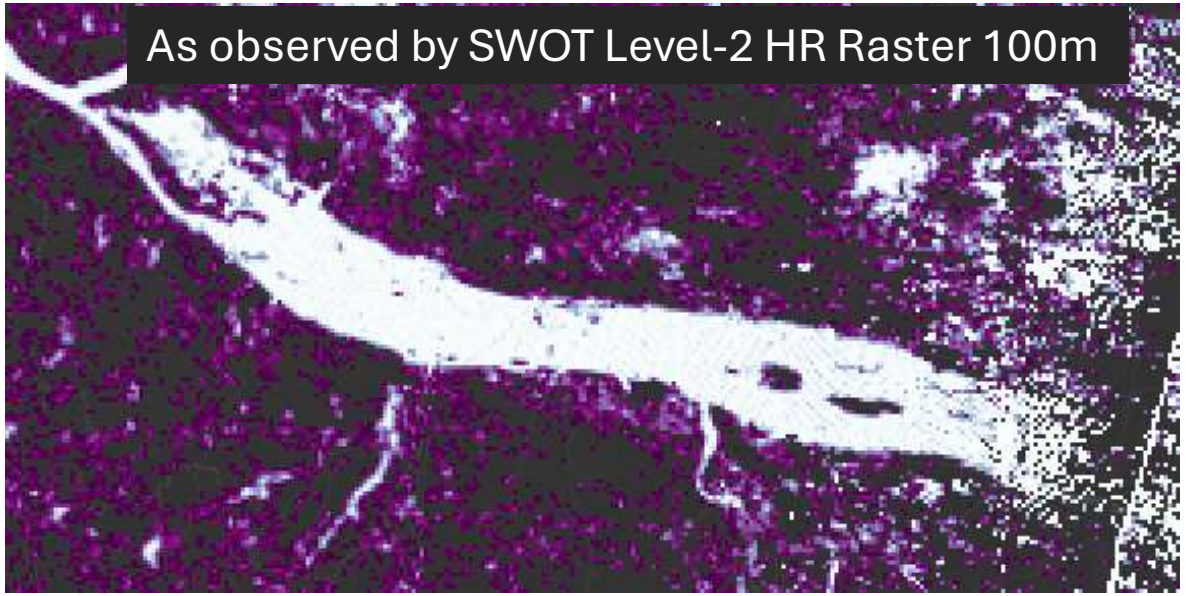
Flood map derived from Sentinel-1 SAR data
Forest and urban classes from ESA World Cover 2021

Context

Yes!

Vienne river flood… on March 31st, 2024





17h49 UTC → Only 2h33 apart! ← 15h16 UTC

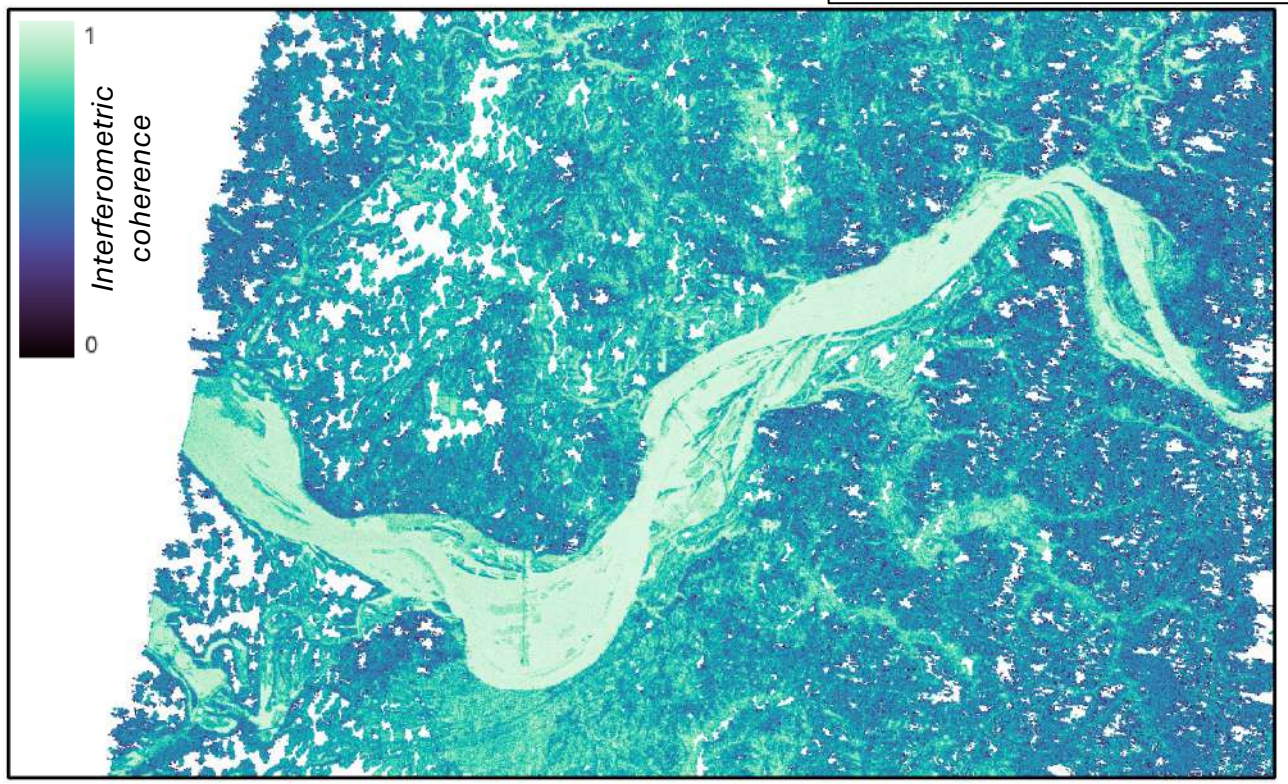
What can SWOT bring to the study of flooded areas compared to side-looking SAR and optical imagery?

Context

Can we get a 2D flood extent of SWOT KaRIn PIXC?

by trying to get the most out of it

Flood seen by SWOT in Ohio river (02-20-2025)

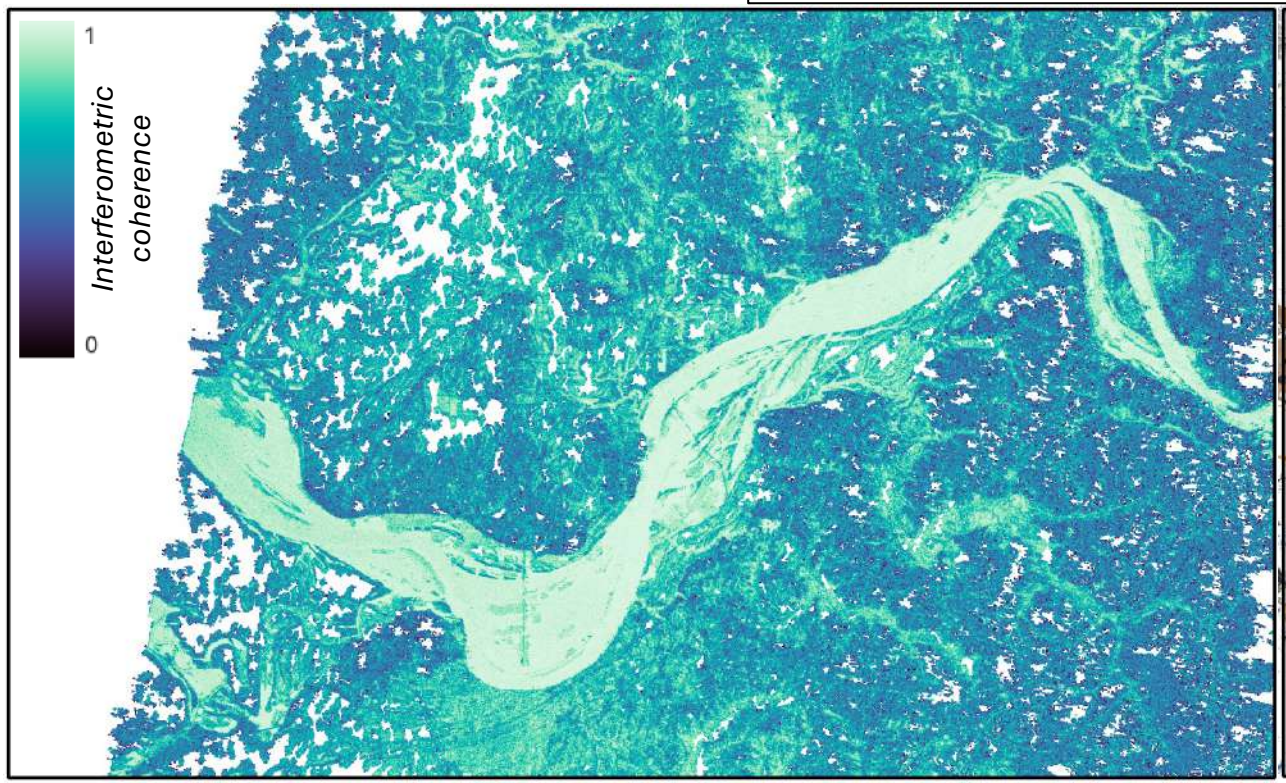


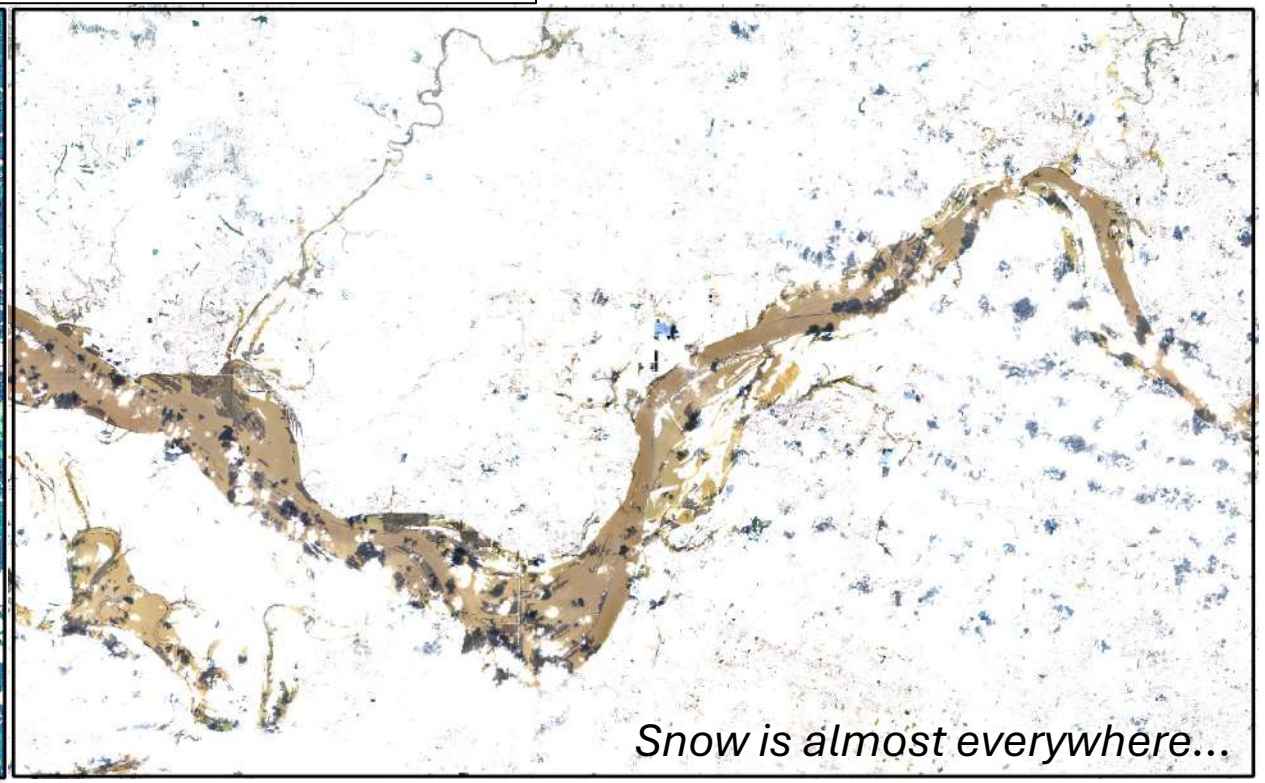
Context

Can we get a 2D flood extent of SWOT KaRIn PIXC?

by trying to get the most out of it

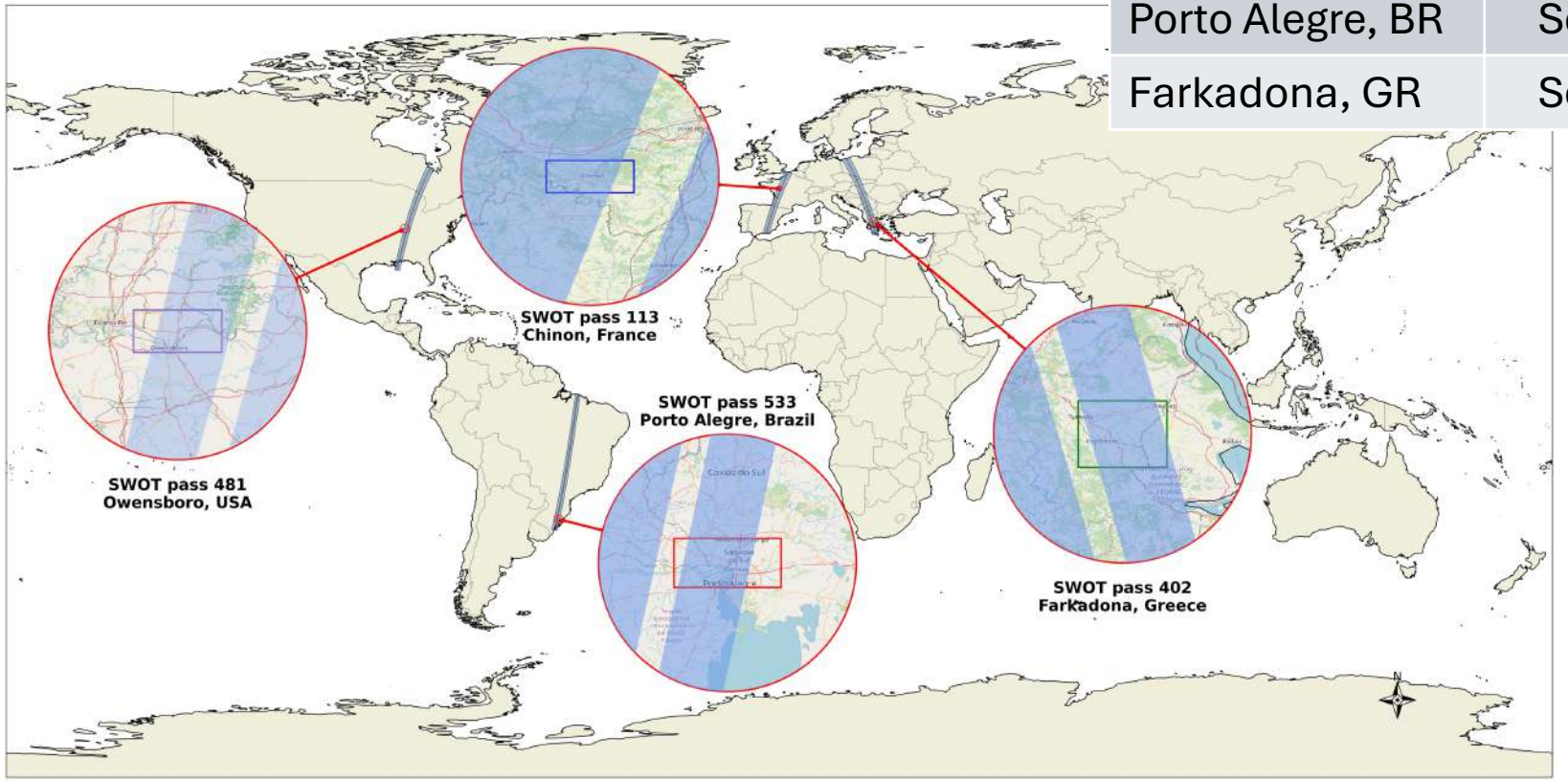
Flood seen by SWOT and S2A in Ohio river (02-20-2025)





Study cases

	Comparing with	Time difference with SWOT
Chinon, FR	Sentinel-1	- 02:32
Owensboro, USA	Sentinel-2	- 01:45
Porto Alegre, BR	Sentinel-2	+ 01:30
Farkadona, GR	Sentinel-2	- 02:12



Product used:

The L2_HR_**PIXC_*** (Pixel Cloud) version C

Many, many variables, among them:

>> “Native” classification of PIXC (water/land, etc.)

>> Backscattering coefficient (sigma0)

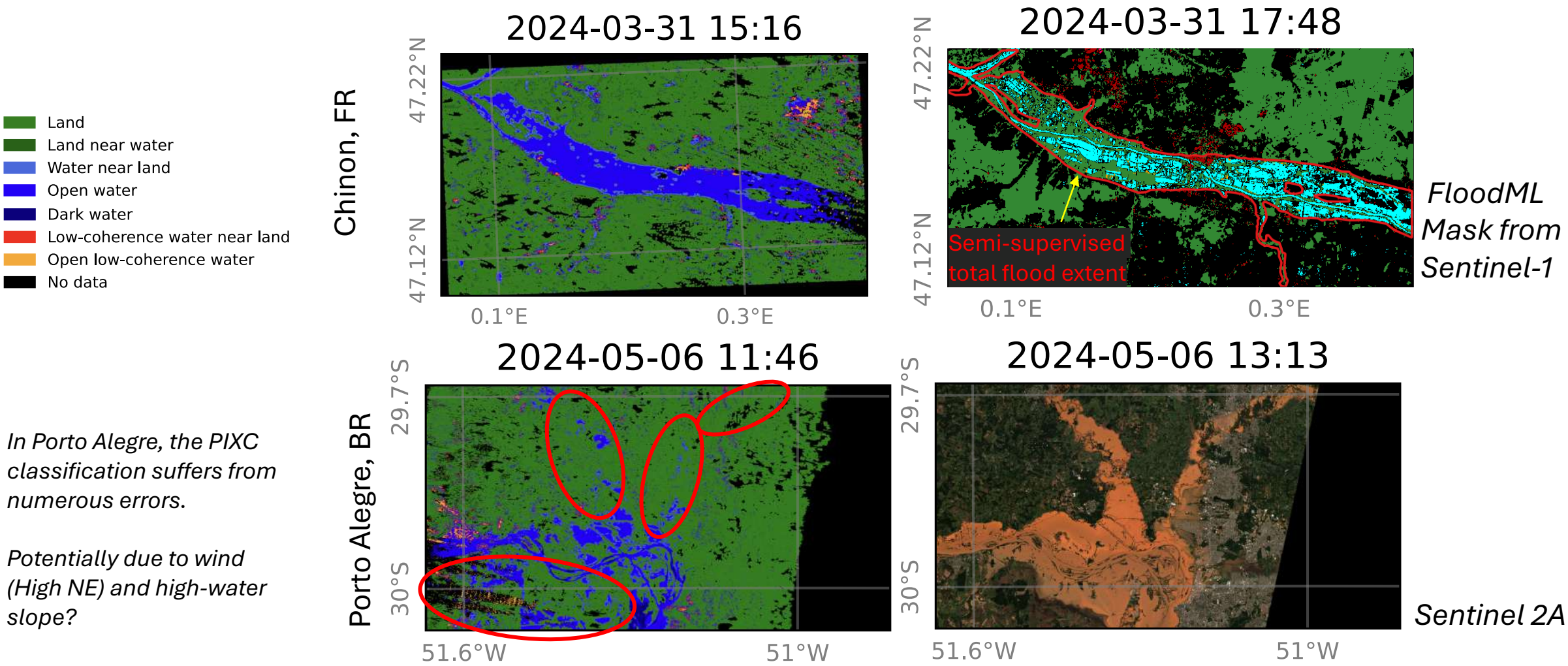
>> Coherent power (P_{coh})

>> Interferometric coherence (γ_{tot})

>> SNR

$$\gamma_{total} = \frac{\langle s_1^* s_2 \rangle}{\sqrt{\langle |s_1|^2 \rangle \langle |s_2|^2 \rangle}}$$

What about the classification?



In Porto Alegre, the PIXC classification suffers from numerous errors.

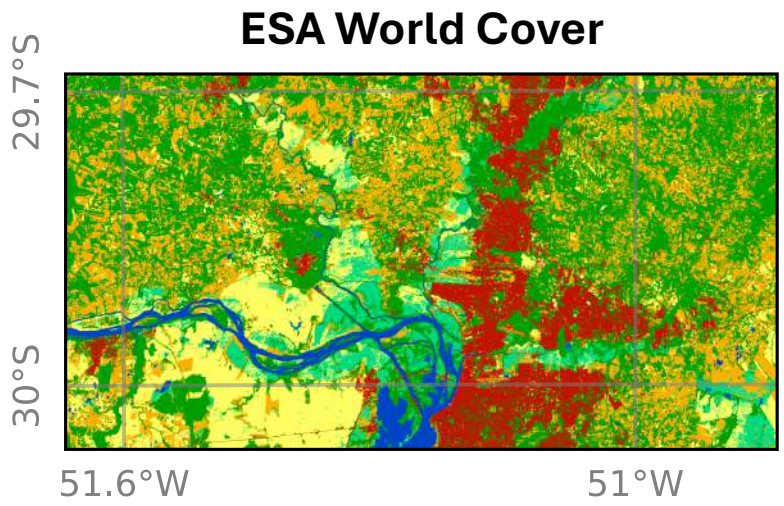
Potentially due to wind (High NE) and high-water slope?

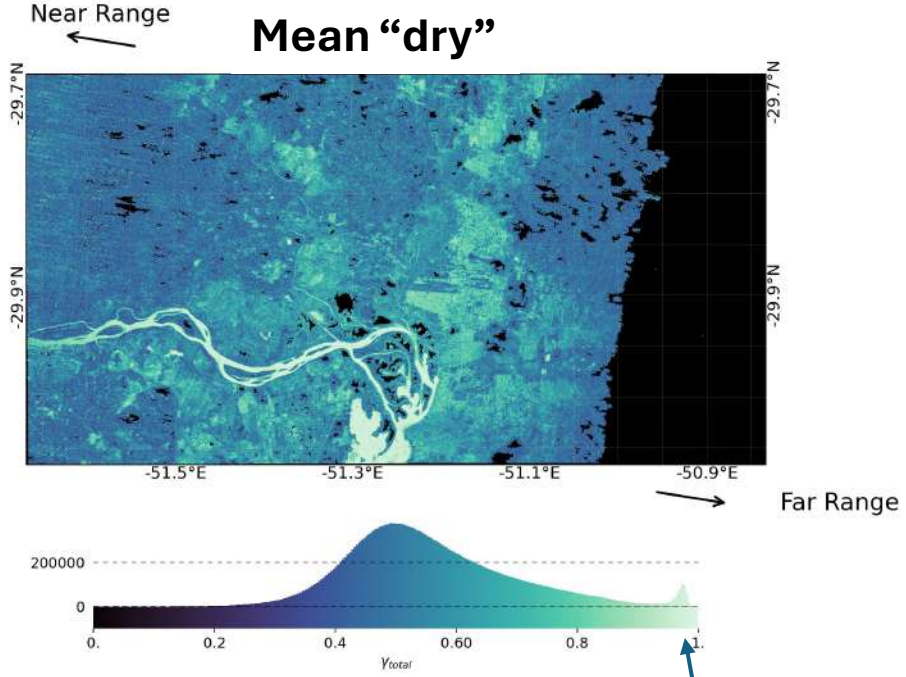
SWOT classification have a lot of uncertainty and can remove quite a lot of water pixel during flood.

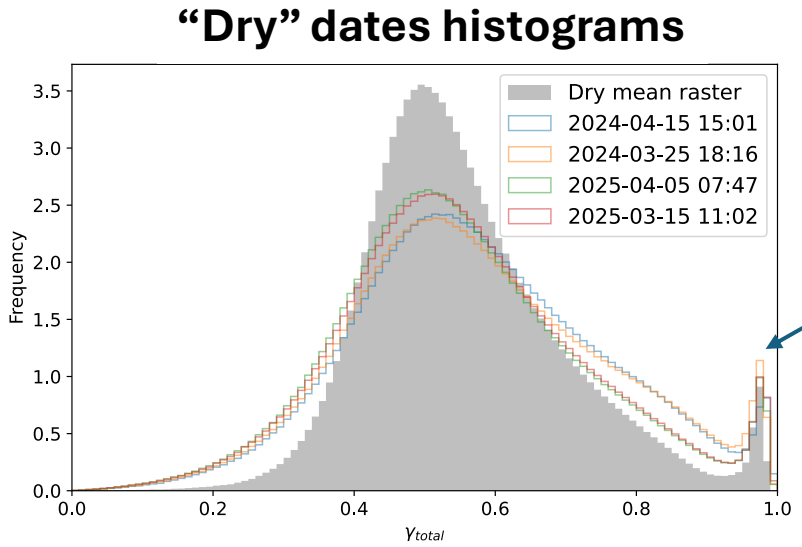
Can we bypass the SWOT classification and use other variables?

Method – Creating Flood map

- Comparing SWOT flood dates with a dry mean raster
- Comparing within three ESA World Cover classes





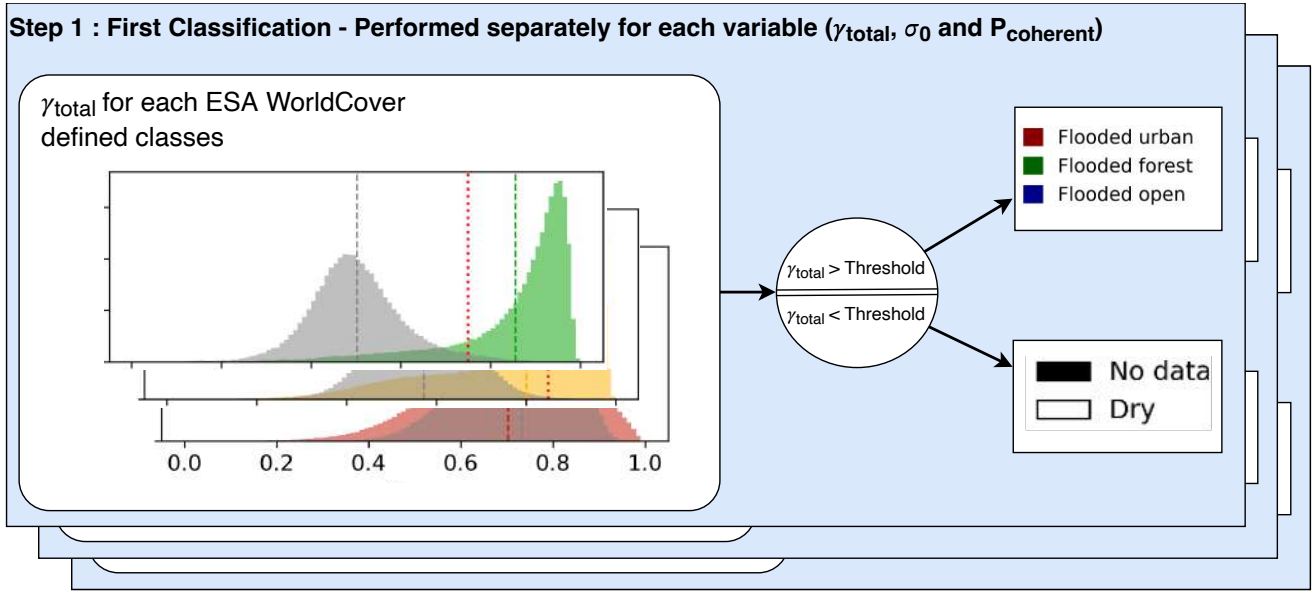


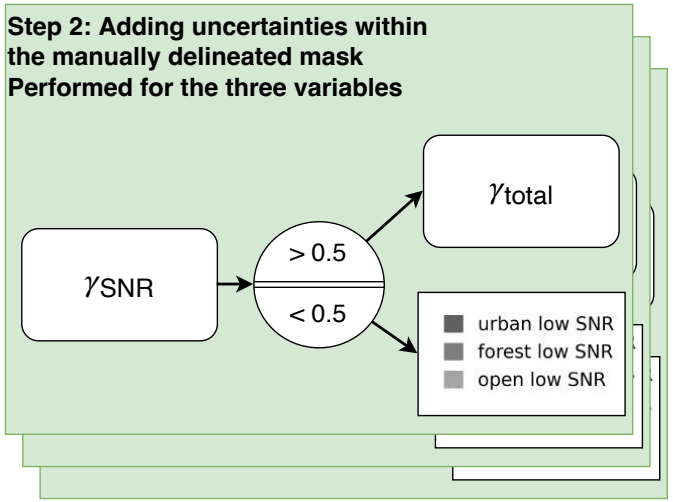
Water peak visible

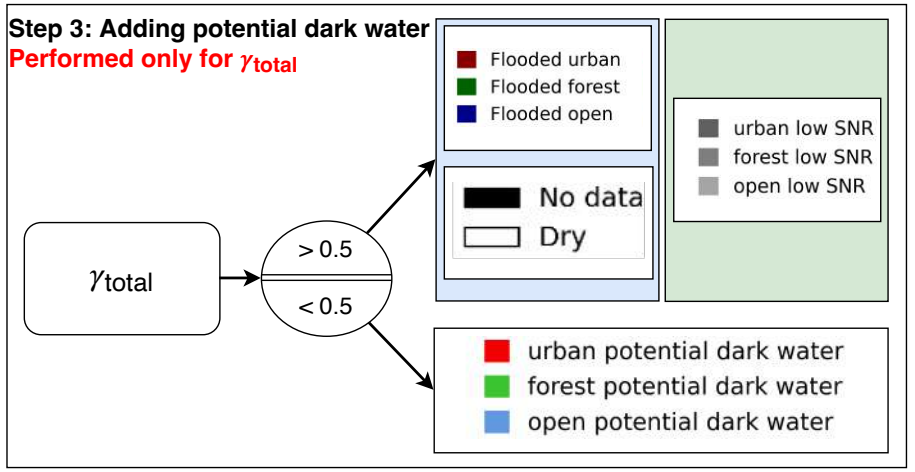
Method – Creating Flood map

- Threshold the variable signal
- Adding uncertainty information within the maps using:
  - γ_{SNR}
  - *a priori* flood map

more details in the preprint…

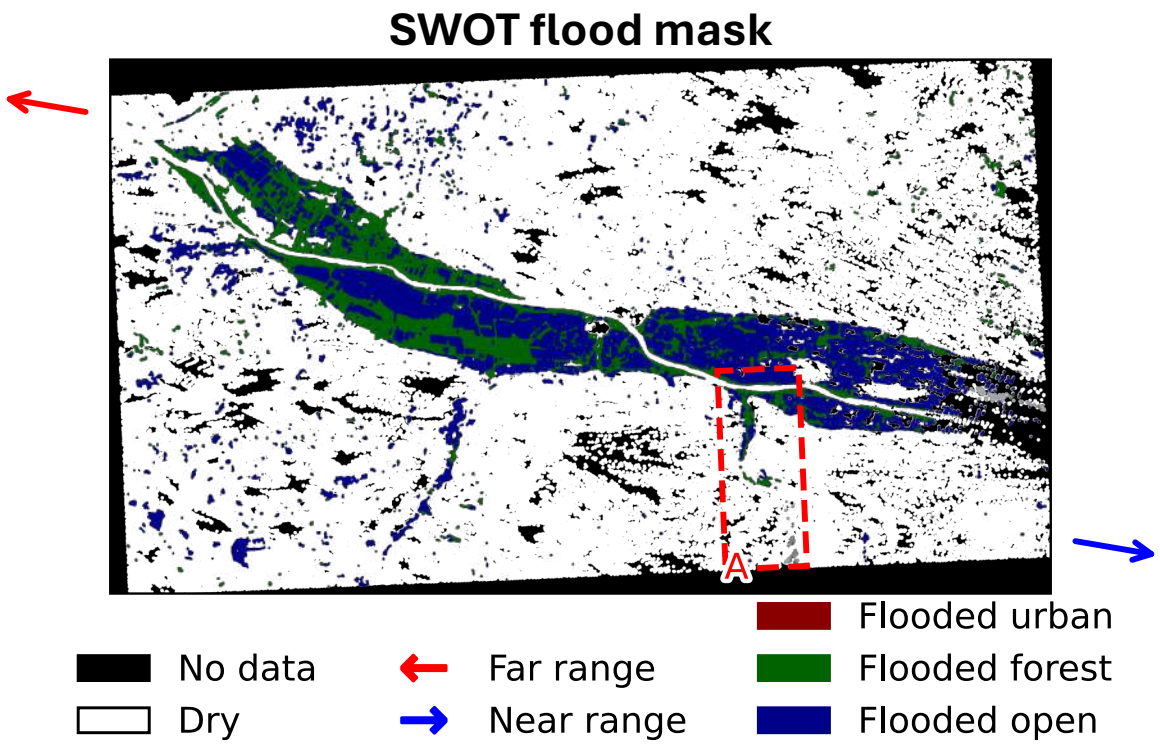


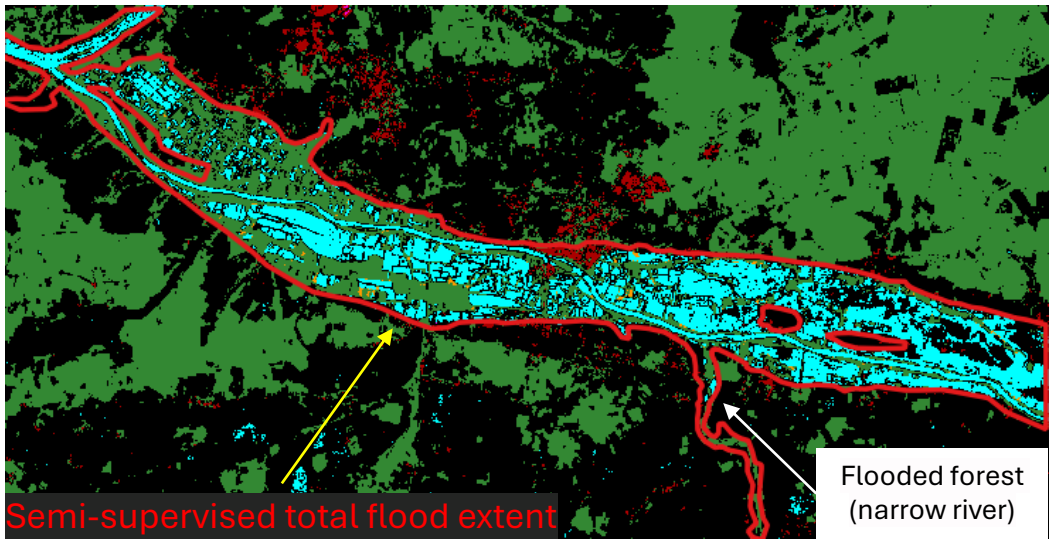




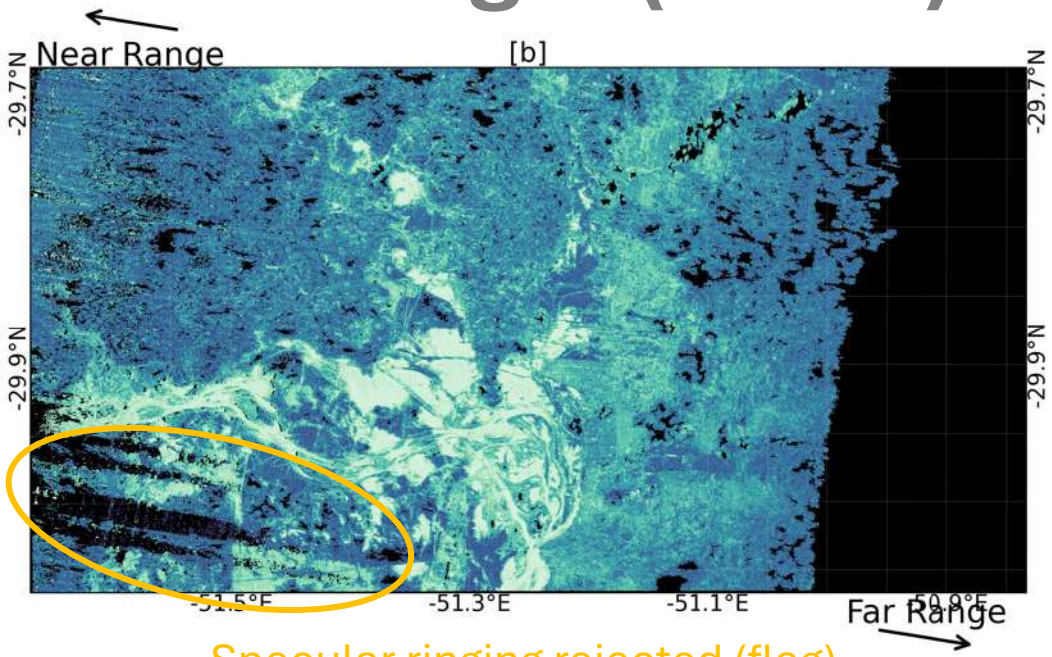
Chinon (France) – Flooded vegetated areas

Interesting new apport: **flooded forest well observed!**



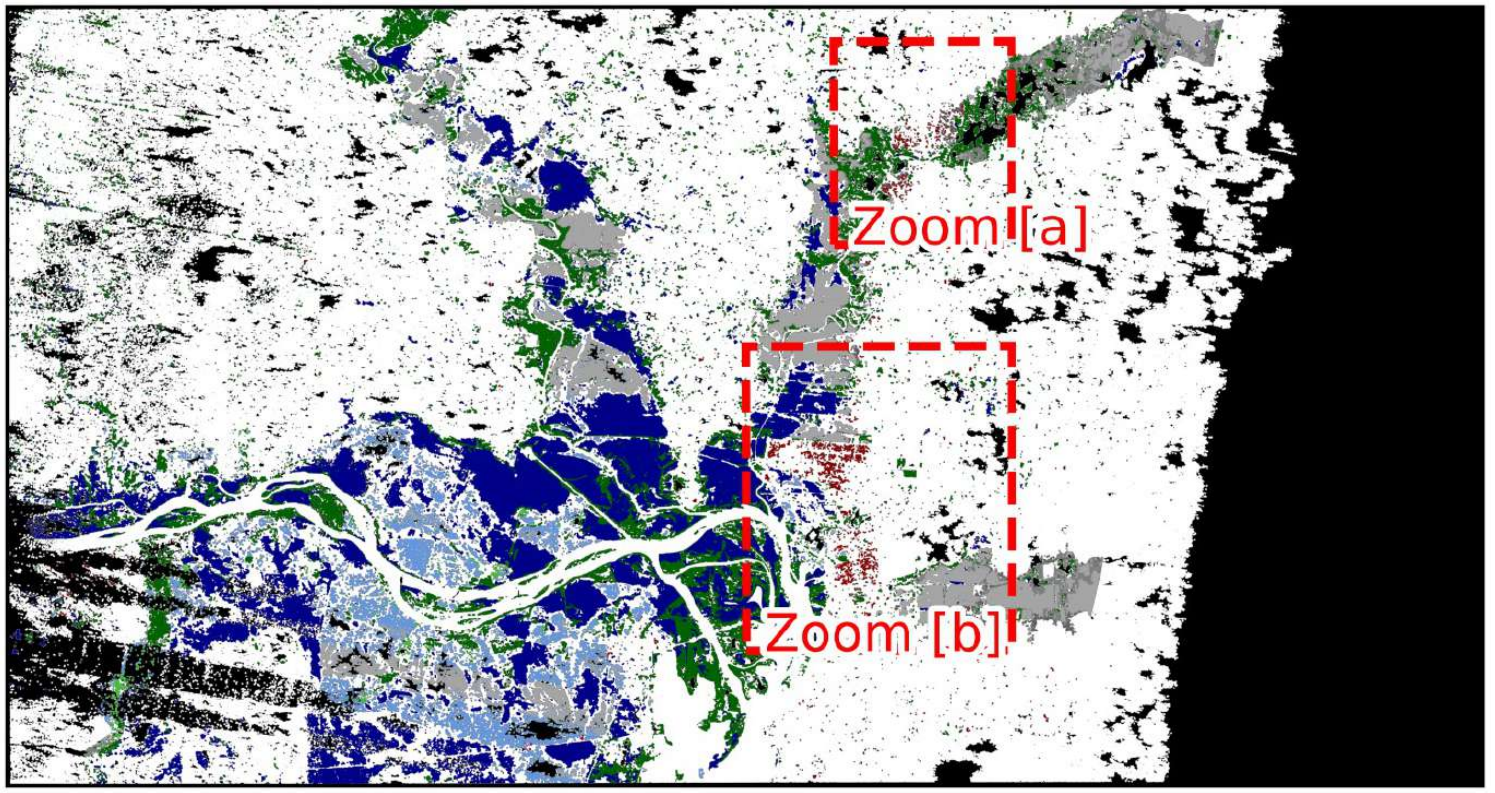


Porto Alegre (Brazil) – Dark-water and urban areas



Specular ringing rejected (flag)

Low coherence + High SNR: potential dark water



Low coherence (dark water)
+ Low SNR

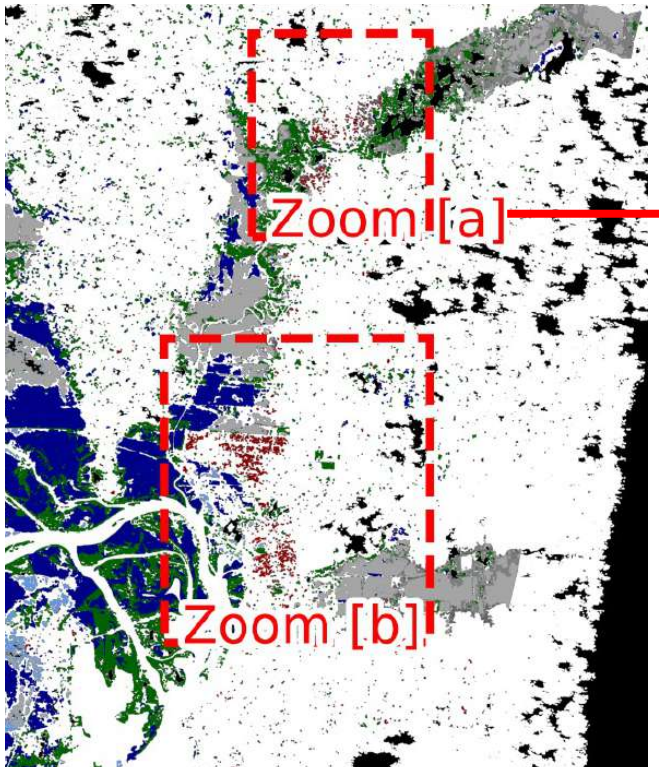
=> excludes unreliable areas

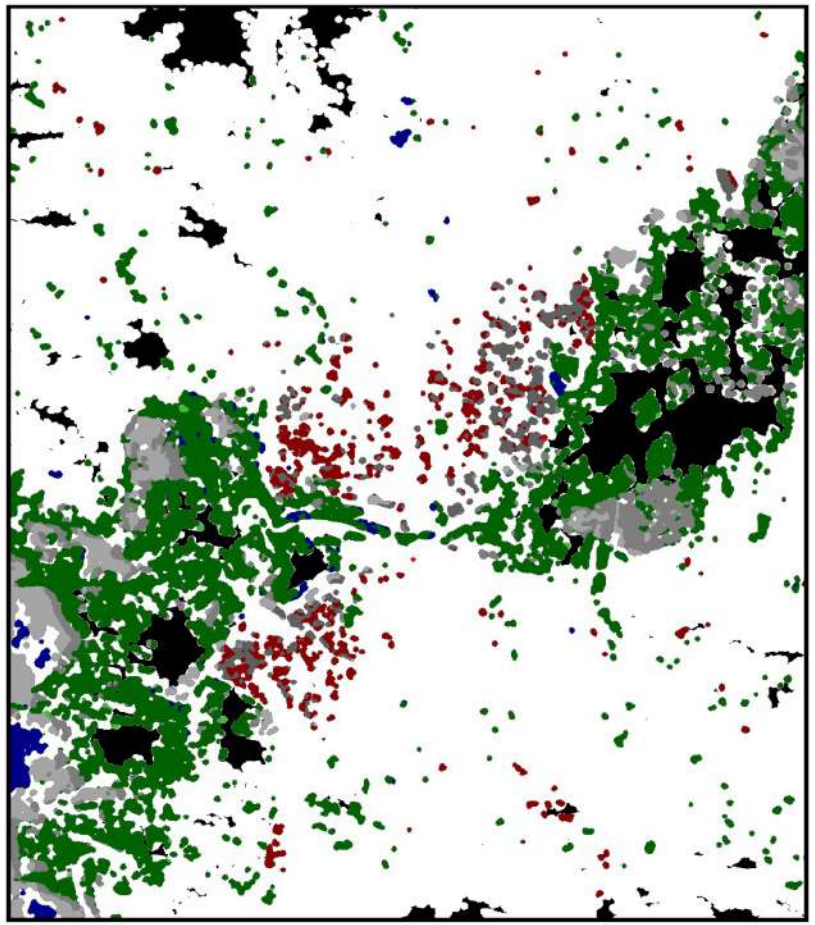
No data
Dry
Flooded urban
Flooded forest
Flooded open
urban potential dark water
forest potential dark water
open potential dark water
urban low SNR
forest low SNR
open low SNR

All AOI

Within Handmade flooded area

Porto Alegre (Brazil) – Dark-water and urban areas



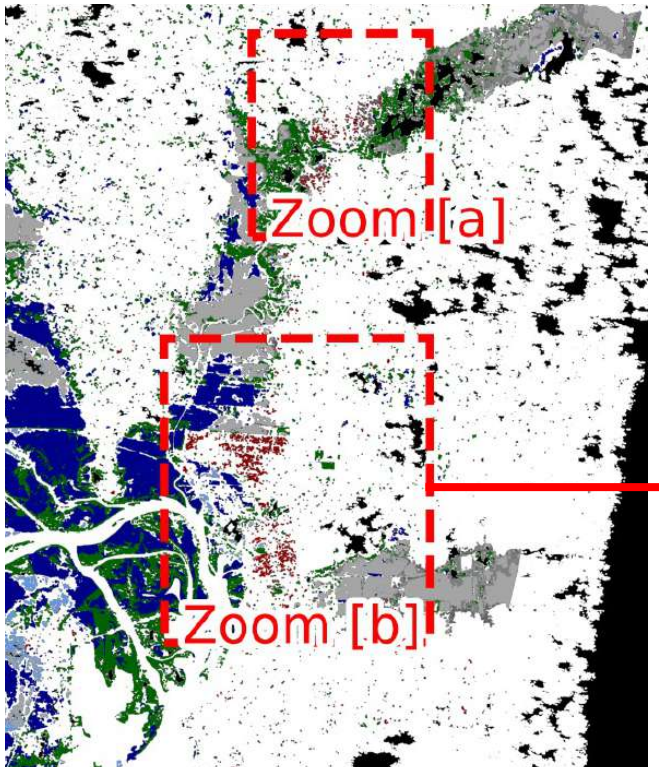


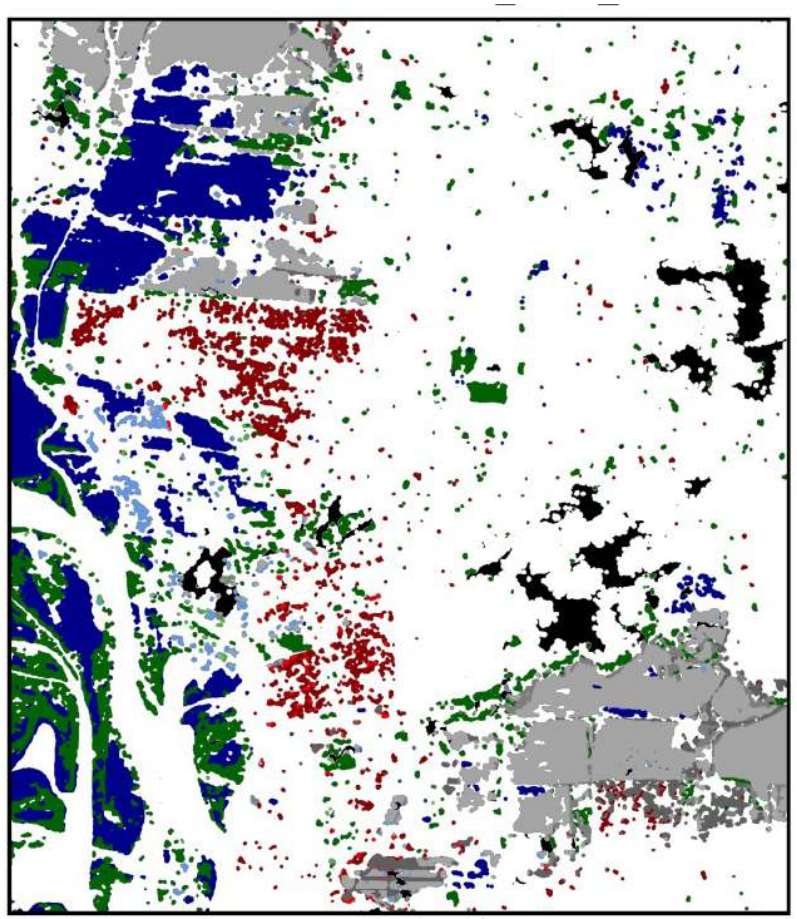


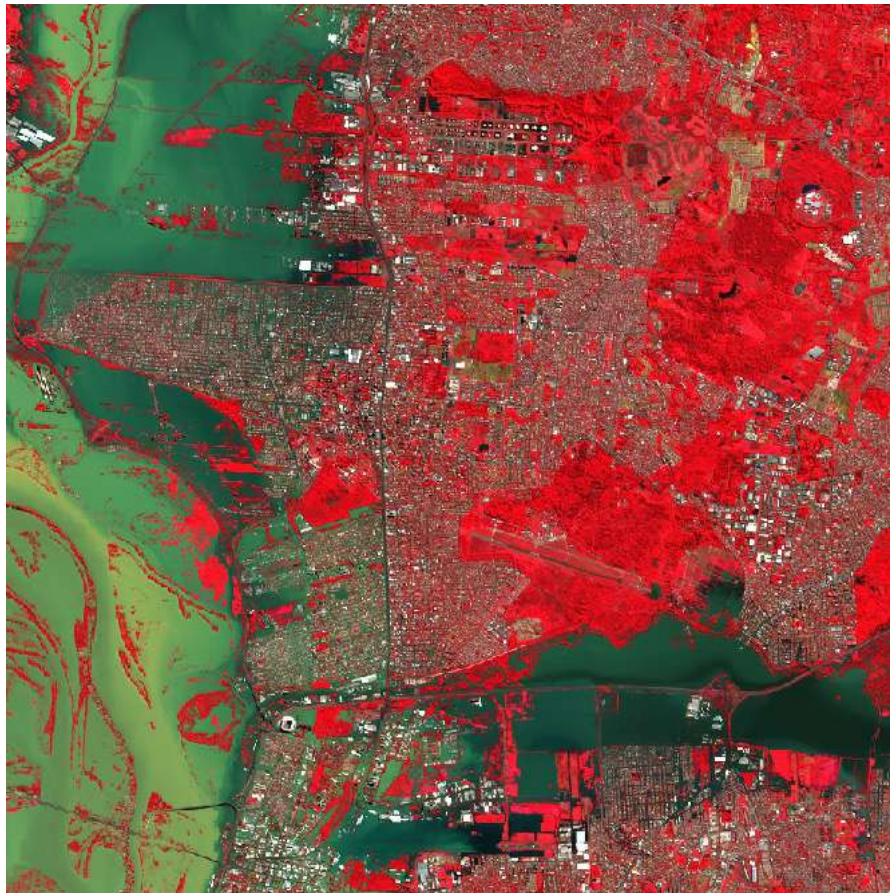
We even detect flooded urban areas directly!

No data
Dry
Flooded urban
Flooded forest
Flooded open
urban potential dark water
forest potential dark water
open potential dark water
urban low SNR
forest low SNR
open low SNR

Porto Alegre (Brazil) – Dark-water and urban areas







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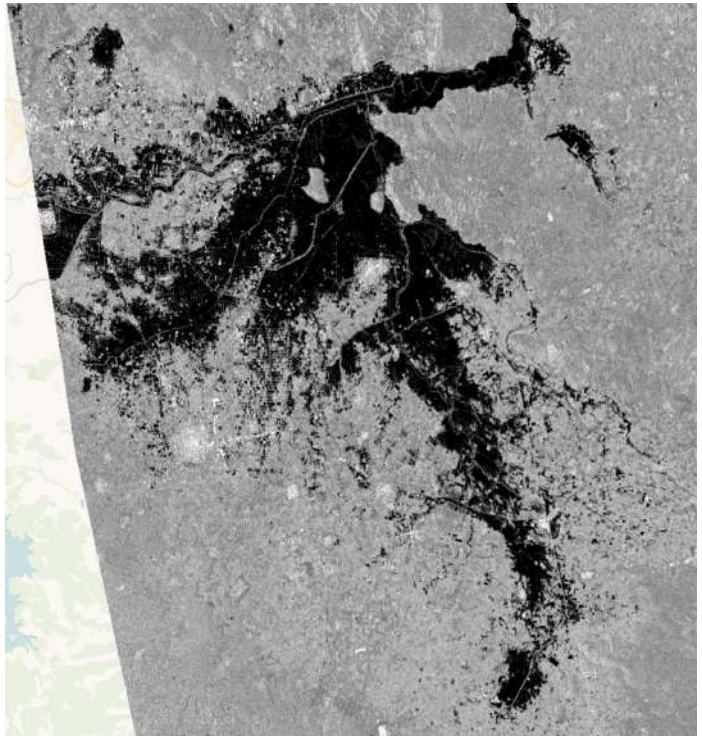
No data
Dry
Flooded urban
Flooded forest
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urban potential dark water
forest potential dark water
open potential dark water
urban low SNR
forest low SNR
open low SNR

Farkadona (Greece) – Flood recession

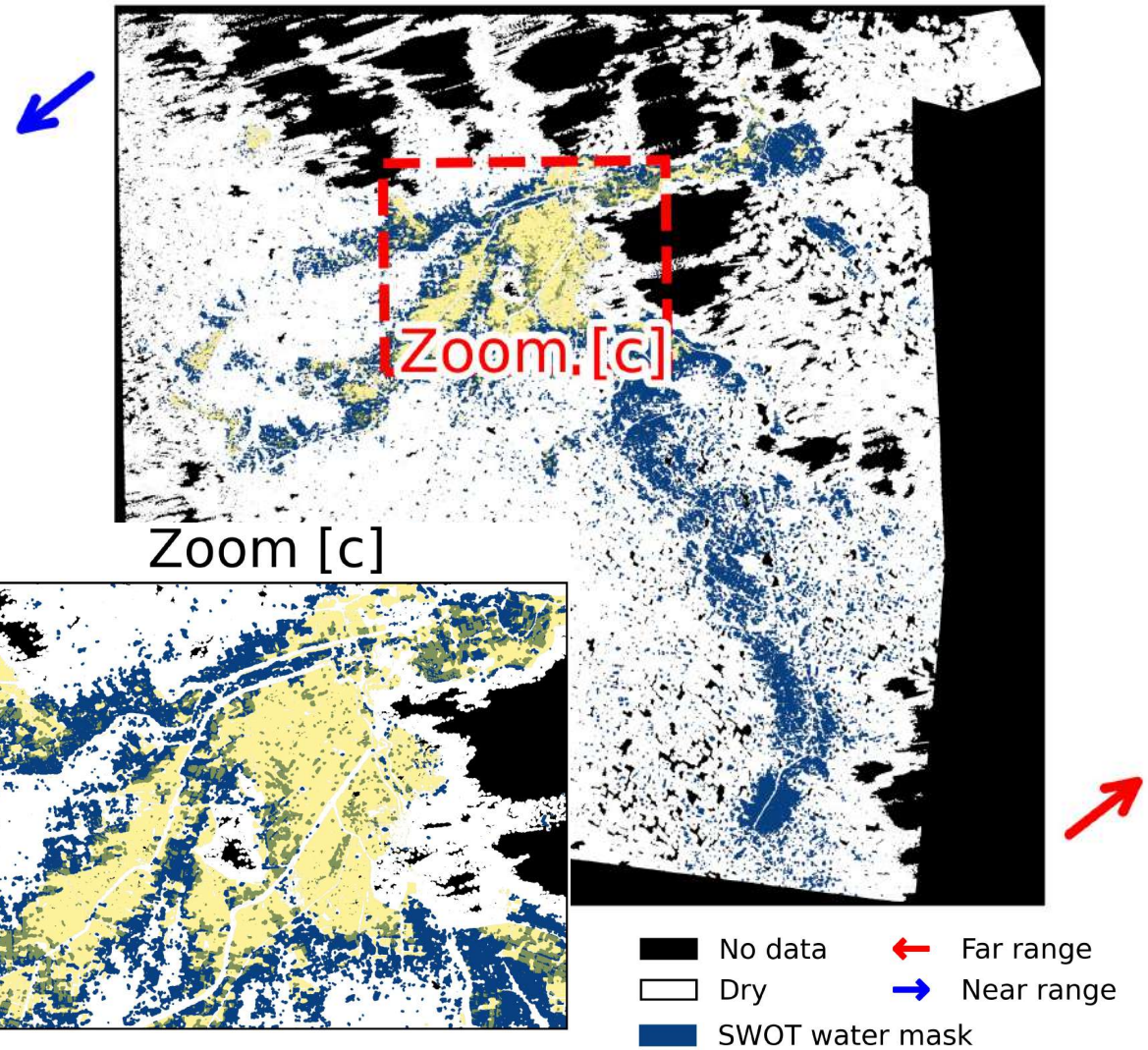
High soil moisture due to flood recession

>> SWOT extent look more like the flood peak extent (few days before)

Sentinel 1 (2023-09-07)



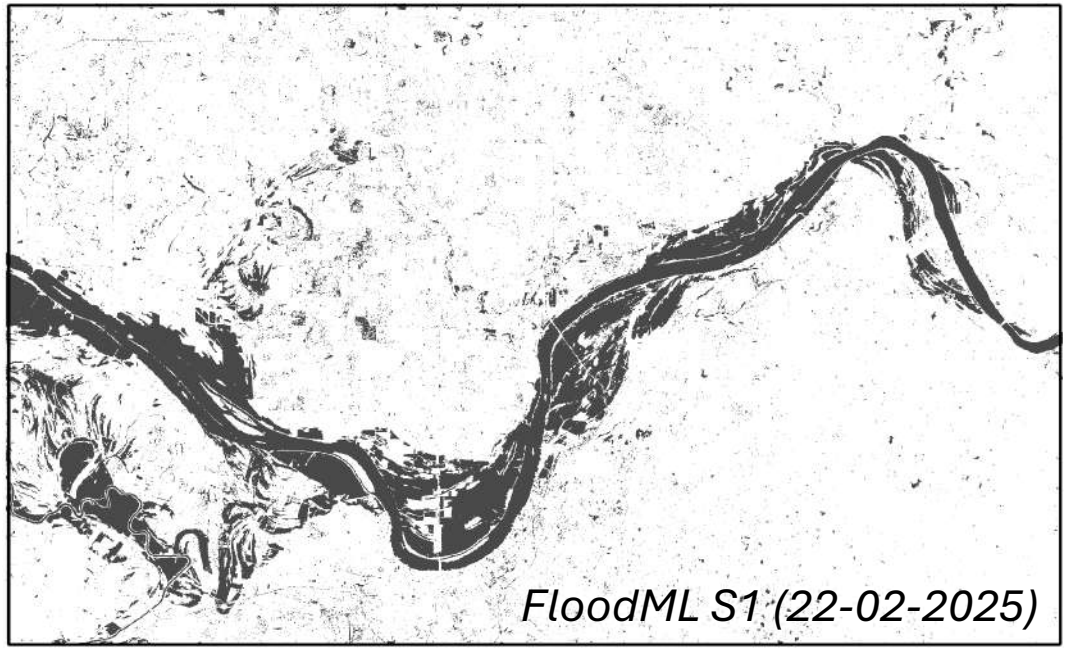
SWOT flood mask (2023-09-15)



Owensboro (USA) – Snow cover

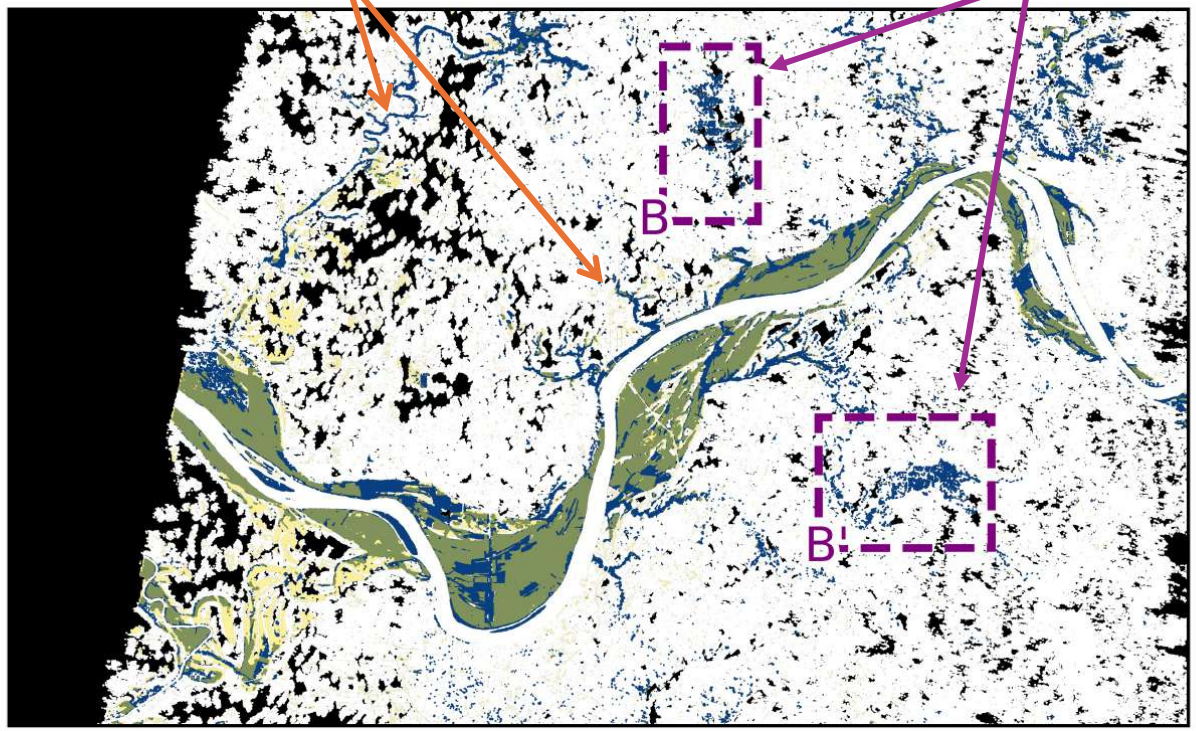
How much does the snow impact the flood extent?

Still, a lot of detail!



Narrow rivers detected

Water under snow, “wet” snow, melting snow



Take-home messages

New techniques to detect flooded areas from SWOT PIXC:

- Flooded forest & open areas (threshold on Sigma0, P_{coh}, γ_{tot})
- Exclusion of specular ringings with flag
- Tackle better dark water areas
  - Current Dark_water flag (within GSWE)
  - Low coherence + High SNR: potential dark water
- ...but also, to detect flooded urban areas!!
  - $\Delta\gamma_{tot}$ between mean dry and flood rasters from PIXC
- Sensible to soil moisture

So many promising things with SWOT... and Sentinel-3 NG?



Thank you !!

Access to preprints, data, github, poster...

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Bonassies et al. 2025 (2026?)
--> In revision
--> Much more in depth analysis

PhD Manuscript in coming